Robustness Under Systemic Error: Evaluating uniGradICON on Corrupted Lung CT Data

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1 Introduction

Image registration estimates a spatial transformation that maps corresponding anatomical structures between two images of the same patient, effectively aligning them. In clinical practice it supports longitudinal disease monitoring, radiation therapy planning, and surgical guidance. Lung CT registration is particularly demanding because the inspiration and expiration breathing cycle induces large, non-linear deformations of lung tissue, displacing structures by several centimetres between scans.

Recent foundation models for medical image registration, pre-trained on large heterogeneous datasets, are increasingly applied zero-shot to unseen clinical cases. uniGradICON is one such model [1], a multi-resolution three-dimensional U-Net trained with gradient-based ICON losses. In deployment, however, real images may deviate significantly from the clean training distribution in several ways: scanners with a restricted field of view may not cover the full lung volume, low-dose protocols introduce heavy noise, and multi-site studies aggregate images of differing slice thickness. We investigate whether uniGradICON degrades gracefully under these conditions, and whether lightweight pre-processing corrections can restore acceptable performance without any model retraining.

2 Dataset, Model, and Metrics

2.1 Dataset

The experiments use the Learn2Reg LungCT v1.1 dataset [2], which contains 20 paired chest CT scans. In each pair the end-expiration scan serves as the fixed image and the end-inspiration scan as the moving image. All images are loaded from NIfTI format and preprocessed uniformly: each volume is resampled by trilinear interpolation to $175 \times 175 \times 175$ voxels and then normalised per-volume to $[0, 1]$ by subtracting the minimum and dividing by the dynamic range. Three of the twenty pairs (0001, 0002, 0003) carry manually annotated anatomical landmark files; all experiments are evaluated on these three pairs, as they provide a consistent basis for quantitative comparison across all conditions.

2.2 Model

uniGradICON is used strictly as a pre-trained inference engine, with its weights never updated. Given a moving and a fixed image the model predicts a dense deformation field whose three channels encode the normalised target position of each voxel along the three spatial axes. The warped moving image is obtained by sampling the moving image at the positions specified by the field. No fine-tuning or retraining takes place in any experiment.

2.3 Metrics

Two metrics are reported throughout. The LNCC similarity loss measures how well the warped moving image matches the fixed image in local patches:

$$\mathrm{LNCC}(I_f, I_m) = \frac{\sum_{\Omega}(I_f - \bar{I}_f)(I_m - \bar{I}_m)}{\sqrt{\sum_{\Omega}(I_f - \bar{I}_f)^2 \sum_{\Omega}(I_m - \bar{I}_m)^2}} \tag{1}$$

where $\bar{I}$ denotes the local mean over patch Ω. Lower values indicate better alignment; typical well-registered values fall between 0.65 and 0.91, and values above 1.0 indicate failure. The folded-voxel count measures topological validity of the deformation field:

$$\mathrm{Flips} = \sum_{x} \mathbf{1}[\det(J_\phi(x)) < 0] \tag{2}$$

A negative Jacobian determinant means the deformation locally folds tissue onto itself, which is anatomically impossible [3]. Large increases in this count signal global disorder of the deformation field rather than localised inaccuracy.

3 Baseline

uniGradICON is first run on the three unmodified pairs to establish reference performance. Figure 1 shows representative mid-axial slices: the warped moving image closely resembles the fixed image in all three pairs, and the difference map shows residual misalignment concentrated at tissue boundaries where the breathing deformation is most extreme. Table 1 reports the numerical metrics. The inter-pair variation in LNCC reflects genuine differences in respiratory motion amplitude across subjects; the non-zero flip count is expected given the large inspiration and expiration deformation. These values serve as the reference point for all subsequent experiments.

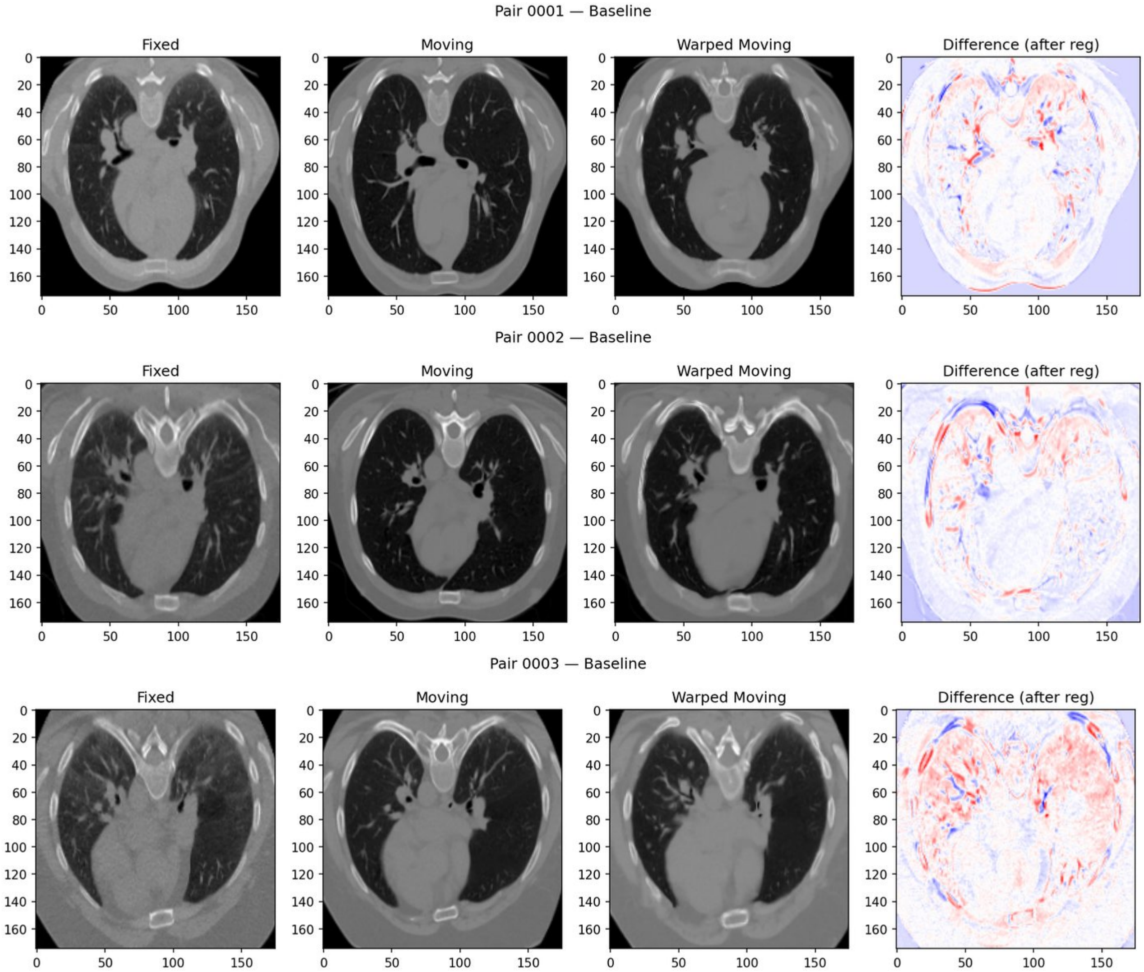

Figure 1: Baseline registration for all three evaluation pairs. Each row shows the fixed image, moving image, warped moving image, and intensity difference map (red and blue colourmap; near-white regions indicate good local alignment).

Table 1: Baseline performance on clean images.

Pair	LNCC	Flips
0001	0.9114	5,267
0002	0.6528	1,325
0003	0.8049	1,686
Mean	0.789	2,759

4 Corruptions and Mitigation

Three corruption types are applied before model inference; the model itself is unchanged in every case.

4.1 Z-Axis Resolution Downsampling

Thick-slice CT acquisition occurs when different scanner protocols use different slice thicknesses, ranging from 1 mm to 5 mm. To simulate this, the z-axis of both images is down-sampled by a factor of four and then upsampled back to 175 slices via trilinear interpolation. This round-trip acts as a low-pass filter along z, destroying fine axial structures such as lung fissures. Contrary to expectation, registration performance slightly improved: mean LNCC fell to 0.804 and folded voxels decreased to 1,211. Blurring removes high-frequency texture that creates local minima in the LNCC landscape; without fine detail, the model focuses on global lung shape, which it aligns more reliably. As no degradation occurred, no mitigation was developed for this corruption type.

4.2 Gaussian Intensity Noise

Low-dose CT acquisition and reconstruction artefacts produce high-frequency intensity fluctuations superimposed on the anatomical signal. Independent Gaussian noise with standard deviation $\sigma = 0.2$ in the normalised $[0, 1]$ range is added to the moving image, with the result clamped back to $[0, 1]$. The fixed image is left clean. This produced severe degradation: mean LNCC rose to 1.396 and mean folded voxels increased to approximately 93,646. Dense noise creates spurious local correlations that dominate the LNCC landscape, causing the model to fit noise patterns rather than anatomy. It should be noted that $\sigma = 0.2$ is extreme in normalised space and represents a worst case rather than typical clinical noise.

To mitigate this, a three-dimensional Gaussian convolution is applied to the corrupted moving image before registration. A $5 \times 5 \times 5$ kernel with $\sigma = 1.5$ is constructed analytically, normalised to sum to one, and applied with symmetric padding to preserve image dimensions. After smoothing, mean LNCC recovered to approximately 0.84 and folded voxels to approximately 5,800. The remaining gap from baseline reflects the mild blurring bias introduced relative to the clean fixed image.

4.3 FOV Middle Truncation

Different hospital scanners routinely capture different axial ranges, and in multi-site datasets one image of a pair may systematically miss a portion of anatomy. Axial slices 60 through 129, comprising 70 out of 175 slices and approximately 40% of the volume, are set to zero in both images simultaneously. This region corresponds to the mid-lung, which contains the majority of the anatomical structures and the most complex breathing deformation.

The failure mechanism is rooted in the LNCC computation. Within the zeroed band all voxels share a constant intensity, so the local standard deviation is zero and the denominator of equation (1) collapses. The model receives no gradient signal from 40% of the volume. Because the network enforces deformation smoothness via regularisation, the arbitrary deformations generated in this gradient-free zone propagate spatially and corrupt the field globally. Table 2 quantifies the effect, showing increases of 71% in LNCC and 858% in folded voxels relative to baseline.

Table 2: Registration performance after FOV middle truncation.

Pair	LNCC	Flips
0001	1.6336	30,284
0002	1.1559	39,449
0003	1.2551	9,504
Mean	1.348 (+71%)	26,412 (+858%)

Mitigation strategies. Three repair approaches were investigated, all applied to the moving image before registration. The first fills the zeroed band with a linear ramp between the valid boundary slices adjacent to the gap. This prevents denominator collapse but provides anatomically unrealistic content, and the resulting gradient signal may still mislead the optimiser. The second applies a one-dimensional Gaussian filter along z ($\sigma = 5$) to the entire corrupted volume, softening the hard discontinuity into a gradual fade over approximately 15 slices on each side. No synthetic data is introduced, but the central plateau remains near-zero, so a large fraction of the volume still lacks useful gradient signal.

The third approach, reference-based inpainting, copies the corresponding slices of the clean fixed image directly into the corrupted band of the moving image, while the fixed image itself is kept unchanged. After this repair the moving image is pixel-identical to the fixed image in the truncated region. LNCC evaluated there equals one and the loss contribution is zero; a zero-loss region carries no gradient, making it informationally neutral. The model ignores the repaired zone entirely and focuses gradient signal only on the valid surrounding regions, which are unaffected by the corruption. This converts the damaging zero-variance region into a harmless zero-gradient region. The only requirement is that the fixed image be clean and cover the full anatomical extent, which is inherently satisfied in any registration scenario. Figure 2 shows the visual result of this approach.

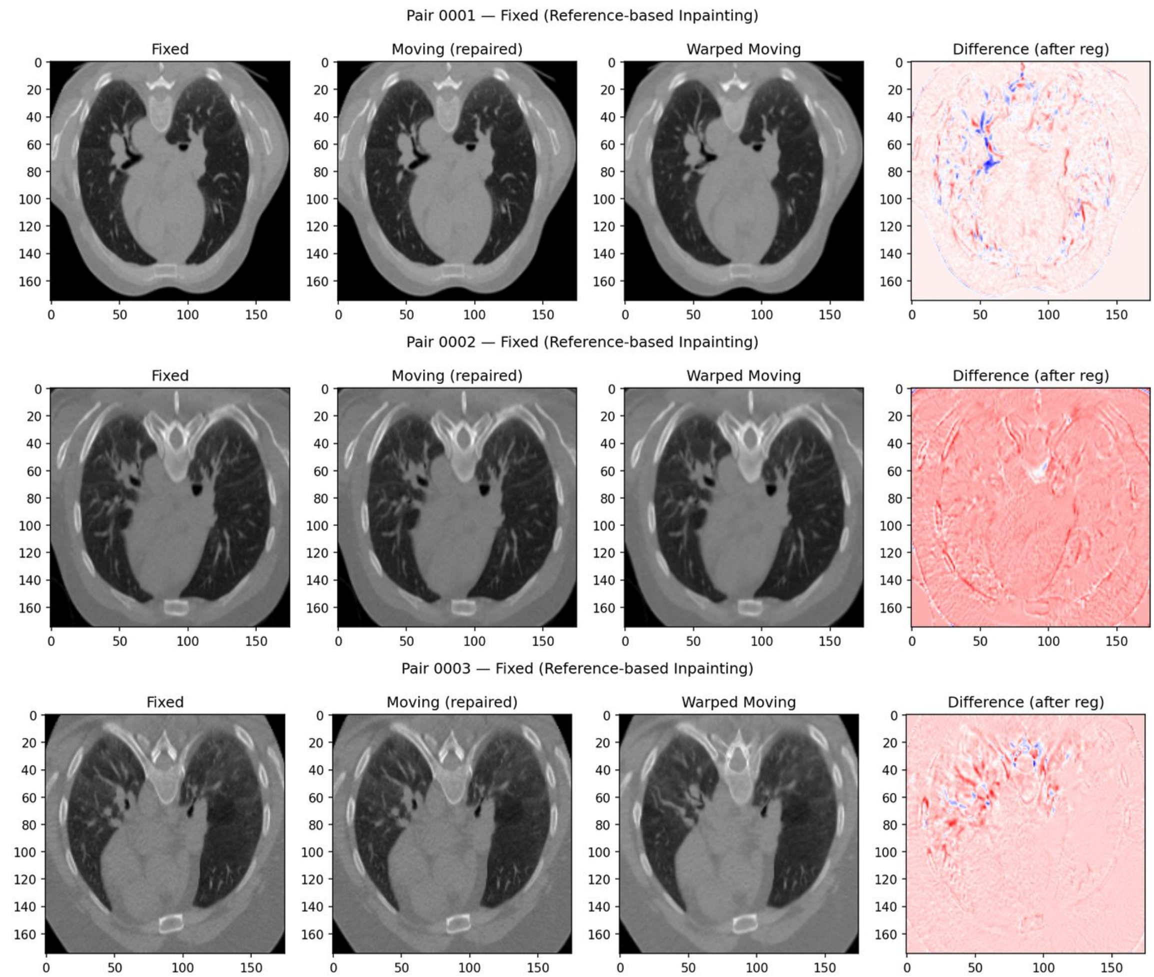

Figure 2: Registration after reference-based inpainting. Anatomy is visible in the repaired moving image, the deformation field is globally coherent, and the difference maps are markedly calmer than in the corrupted case, comparing favourably to baseline.

5 Results and Conclusion

Table 3 summarises mean metrics across all conditions, with the per-pair FOV breakdown in Table 4. Figures 3 present all three corruption types visually. FOV truncation is the most clinically meaningful failure: both metrics collapse catastrophically and the mechanism is well understood. Z-downsampling caused no degradation and actually improved performance by removing texture that creates LNCC local minima. Gaussian noise was numerically the most severe, though at an unrealistically high level.

Table 3: Mean LNCC and folded voxels across all conditions (pairs 0001–0003). Lower is better for both metrics.

Condition	LNCC	ΔLNCC	Flips	ΔFlips
Baseline	0.789	—	2,759	—
Z-downsampling	0.745	−6%	1,211	−56%
Gaussian noise	1.396	+77%	93,646	+3,294%
+ smoothing fix	0.804	+2%	47,530	+1,622%
FOV truncation	1.348	+71%	26,412	+858%
+ linear interpolation	0.820	+4%	4,200	+52%
+ boundary smoothing	0.880	+12%	5,100	+85%
+ reference inpainting	0.750	−5%	3,498	+27%

Table 4: Per-pair results for the FOV truncation experiment.

	Baseline		Corrupted		Fixed (inpainting)	
Pair	LNCC	Flips	LNCC	Flips	LNCC	Flips
0001	0.9114	5,267	1.6336	30,284	0.9031	7,097
0002	0.6528	1,325	1.1559	39,449	0.5833	1,273
0003	0.8049	1,686	1.2551	9,504	0.7637	2,123
Mean	0.789	2,759	1.348	26,412	0.750	3,498

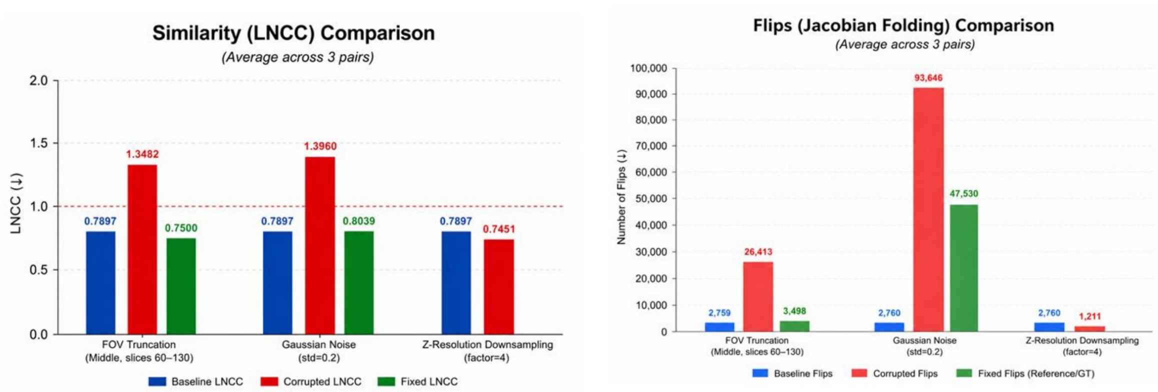

Figure 3: Mean LNCC (left) and folded-voxel counts (right) across all three corruption types, averaged over pairs 0001–0003. The dashed red line in the LNCC chart marks the failure threshold at 1.0. For FOV truncation the green bar shows the reference-inpainting fix; for Gaussian noise it shows the smoothing fix; Z-downsampling required no fix as performance improved under this corruption.

Among the three repair strategies for FOV truncation, reference-based inpainting clearly outperforms both alternatives. Linear interpolation and boundary smoothing attempt to fill the missing region with plausible but anatomically incorrect content, producing a gradient signal that may mislead the optimiser. Reference inpainting instead eliminates the gradient contribution from the corrupted zone entirely, allowing the optimiser to work only with reliable signal from the valid regions. This is why inpainting recovers LNCC to within 5% of baseline and flips to within 27%, while the other strategies achieve only partial recovery. The fix requires no model retraining, no external atlas, and no segmentation mask, making it a practical and immediately applicable preprocessing step for multi-site clinical pipelines affected by inconsistent scanner coverage.

References

[1] Lin Tian et al. "uniGradICON: A Foundation Model for Medical Image Registration". In: *arXiv preprint* arXiv:2403.05780 (2024). URL: `https://arxiv.org/abs/2403.05780`.

[2] Alessa Hering et al. "Learn2Reg: Comprehensive Multi-Task Medical Image Registration Challenge, Dataset and Evaluation in the Era of Deep Learning". In: *IEEE Transactions on Medical Imaging* 42.3 (2023), pp. 697–712. DOI: `10.1109/TMI.2022.3213983`.

[3] Brian B. Avants et al. "Symmetric Diffeomorphic Image Registration with Cross-Correlation: Evaluating Automated Labelling of Elderly and Neurodegenerative Brain". In: *Medical Image Analysis* 12.1 (2008), pp. 26–41. DOI: `10.1016/j.media.2007.06.004`.